R-CNN

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Test

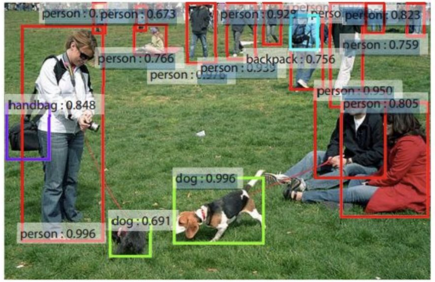
1. Почему при хранении картинки мы используем 3 канала?
2. Какая проблема есть у объектива с одной линзой?
3. Назовите три важнейших цветовых пространства
4. Назовите характерные проблемы Rolling Shutter камер
5. Назовите три библиотеки для работы с изображениями в Пионе (не нейросетевые).

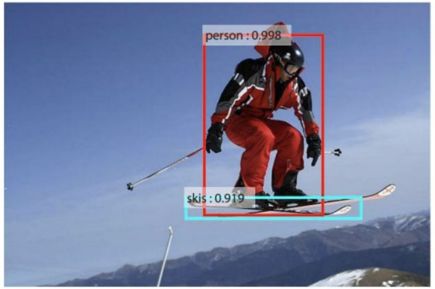
Outline

- Computer Vision tasks
  - Classification
  - Localisation
  - Detection
  - Semantic Segmentation
  - Instance Segmentation
- Metrics
- Datasets
- R-CNN
- Fast R-CNN
- Faster R-CNN

Computer Vision tasks

Computer Vision tasks





- Classification
- Localisation
- Detection

- Semantic Segmentation
- Instance Segmentation

- Style Transfer
- Adversarial Attacks

- Human Pose Estimation
- Person Re-Identification
- Object Tracking
- ………

Classification

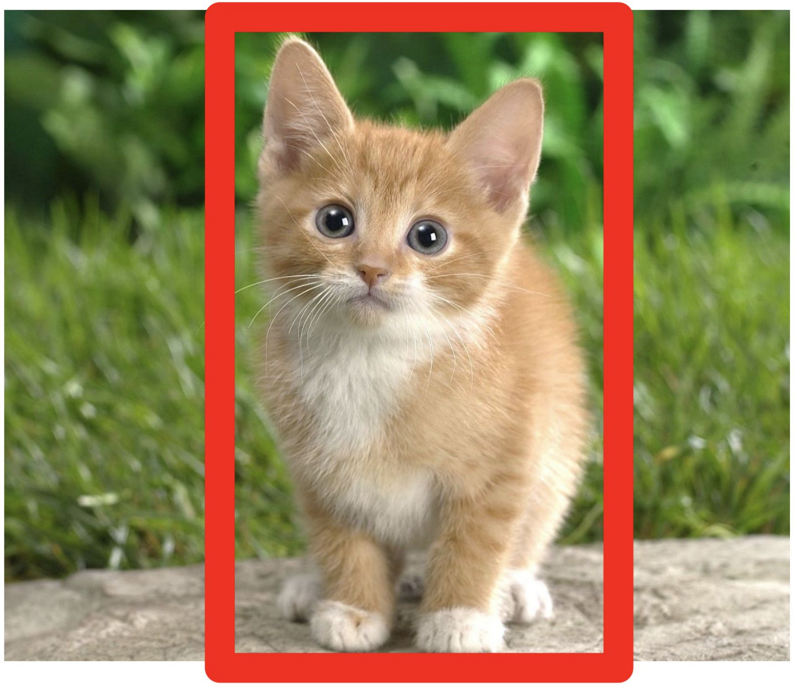
Cat

Input image (C x H x W)

Output

class label (int)

Localisation



Cat

Input image (C x H x W)

Output

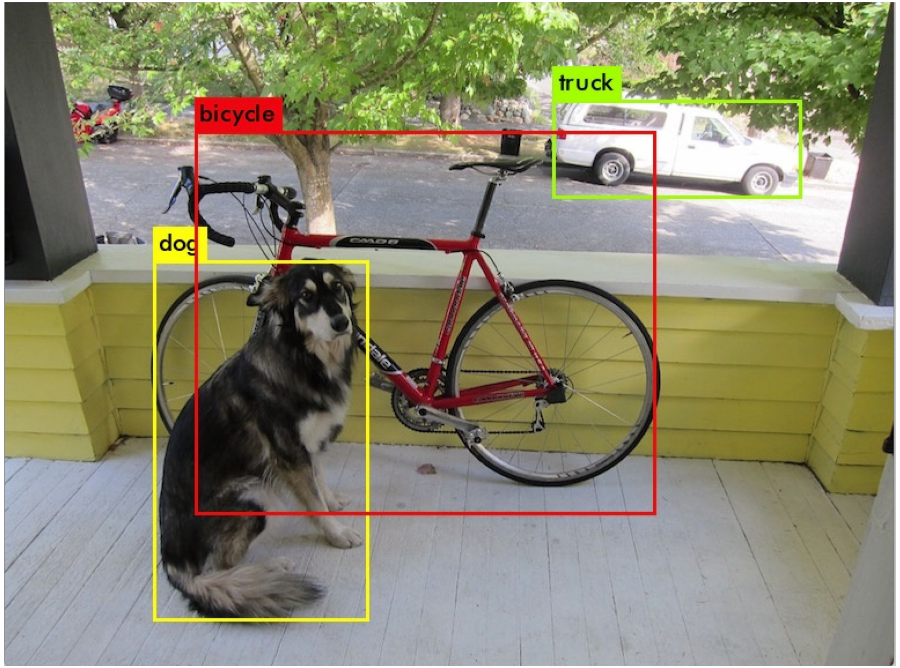
class label (int), BB* coordinates (4 ints)

Assumption

one object per image

* BB - bounding box

Detection



Input image (C x H x W)

Output

class label (int), confidence (0-1 float), BB (4 ints)

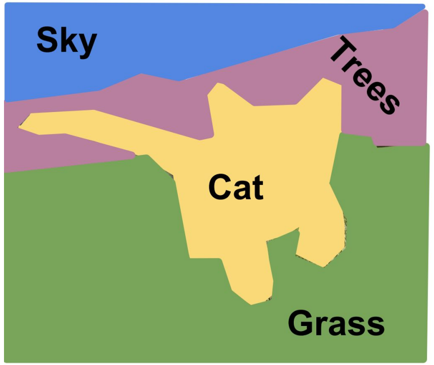
……………….

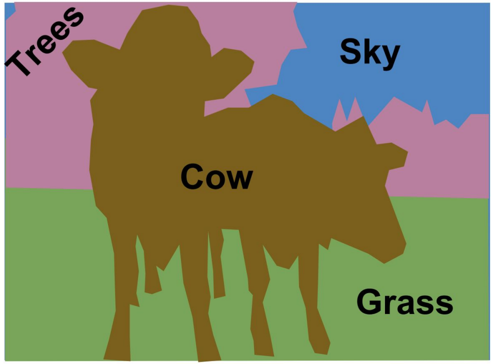
class label (int), confidence (0-1 float), BB (4 ints)

Assumption

multiple objects per image

Semantic Segmentation





Input image (C x H x W)

Output

mask of classes (1 x H x W)

Assumption

just class label for each pixel

Instance Segmentation

Input image (C x H x W)

Output

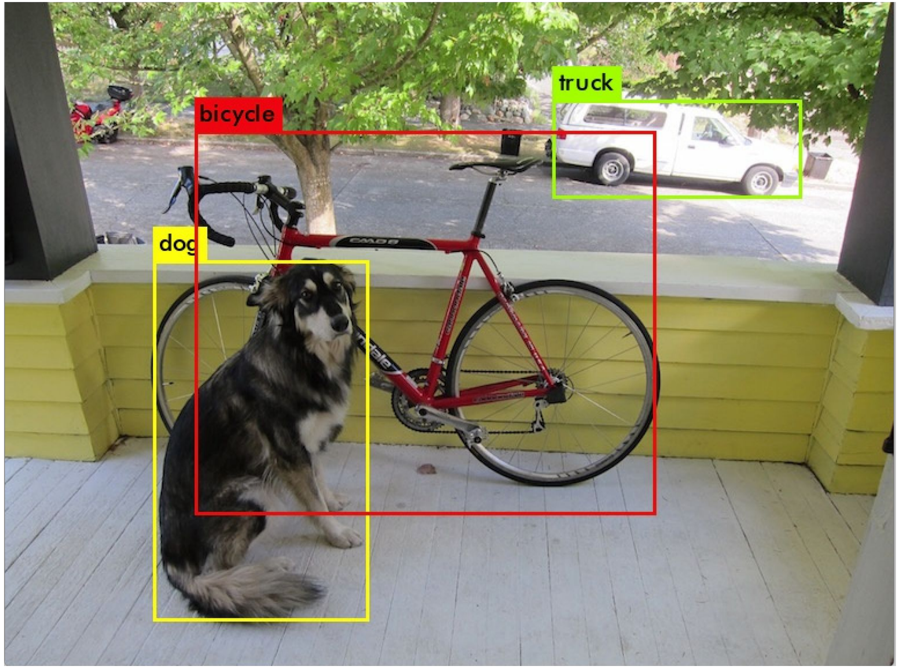
mask of class instances (detection + 1 x H x W)

Assumption

separated masks for each distinct object

Detection metrics

Detection



Input image (C x H x W)

Output

class label (int), confidence (0-1 float), BB (4 ints)

………………..

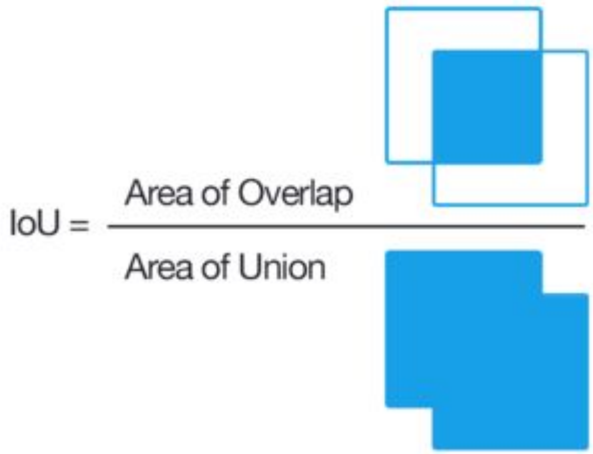
class label (int), confidence (0-1 float), BB (4 ints)

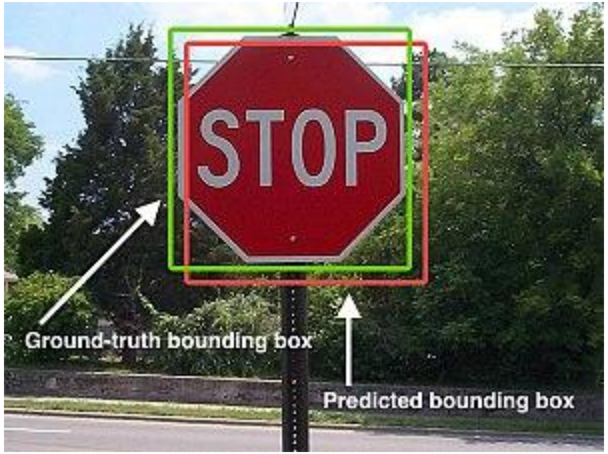
Assumption

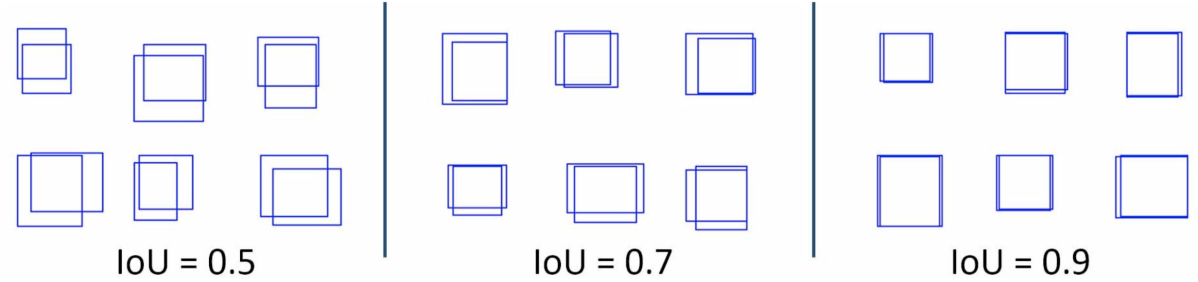
multiple objects per image

Intersection over Union (IoU)

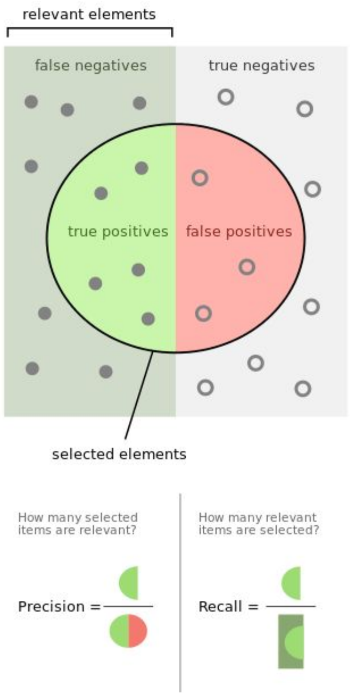
aka Jaccard index







mean Average Precision (mAP)



Consider 5 objects and 10 detections

Rank	Correct?	Precision	Recall
1	True	1.0	0.2
2	True	1.0	0.4
3	False	0.67	0.4
4	False	0.5	0.4
5	False	0.4	0.4
6	True	0.5	0.6
7	True	0.57	0.8
8	False	0.5	0.8
9	False	0.44	0.8
10	True	0.5	1.0

mean Average Precision (mAP)

The intention in interpolating the precision/recall curve in this way is to reduce the impact of the “wiggles” in the precision/recall curve, caused by small variations in the ranking of examples.

Rank	Correct?	Precision	Recall
1	True	1.0	0.2
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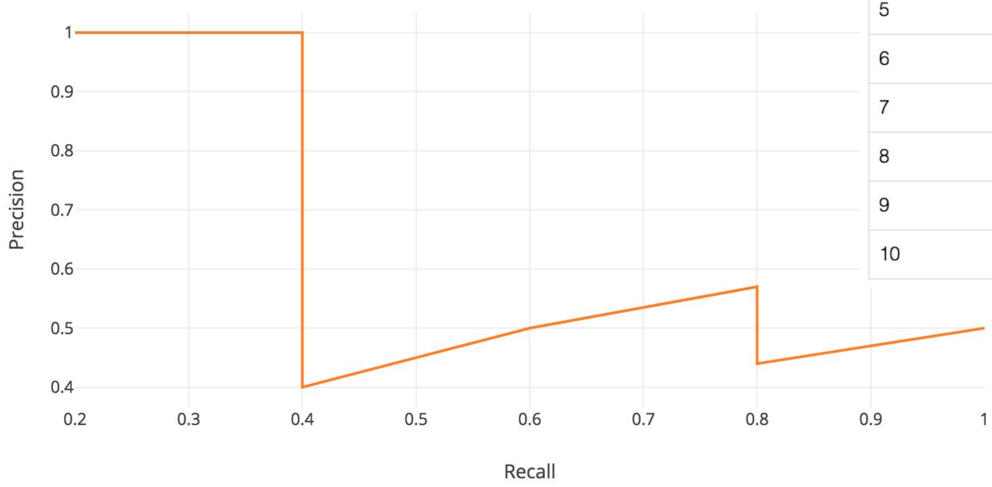
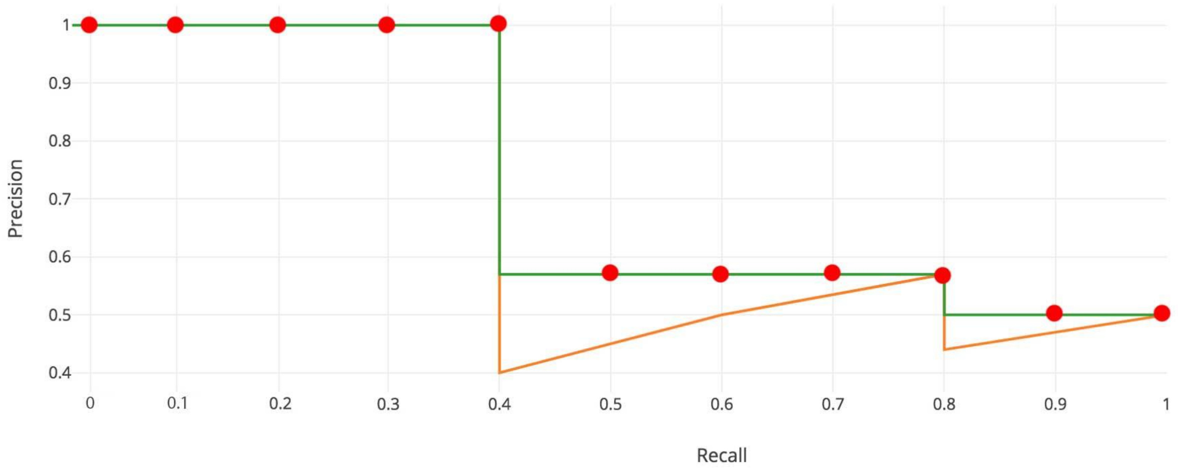


mean Average Precision (mAP)

mAP is mean of AP over all classes

$$AP = \frac{1}{11} \times \left(AP_r(0) + AP_r(0.1) + \ldots + AP_r(1.0)\right)$$



COCO mAP

Average Precision (AP):

AP	% AP at IoU=.50:.05:.95 **(primary challenge metric)**
$AP^{IoU=.50}$	% AP at IoU=.50 (PASCAL VOC metric)
$AP^{IoU=.75}$	% AP at IoU=.75 (strict metric)

AP Across Scales:

AP^{small}	% AP for small objects: area < 32^2
AP^{medium}	% AP for medium objects: 32^2 < area < 96^2
AP^{large}	% AP for large objects: area > 96^2

Average Recall (AR):

$AR^{max=1}$	% AR given 1 detection per image
$AR^{max=10}$	% AR given 10 detections per image
$AR^{max=100}$	% AR given 100 detections per image

AR Across Scales:

AR^{small}	% AR for small objects: area < 32^2
AR^{medium}	% AR for medium objects: 32^2 < area < 96^2
AR^{large}	% AR for large objects: area > 96^2

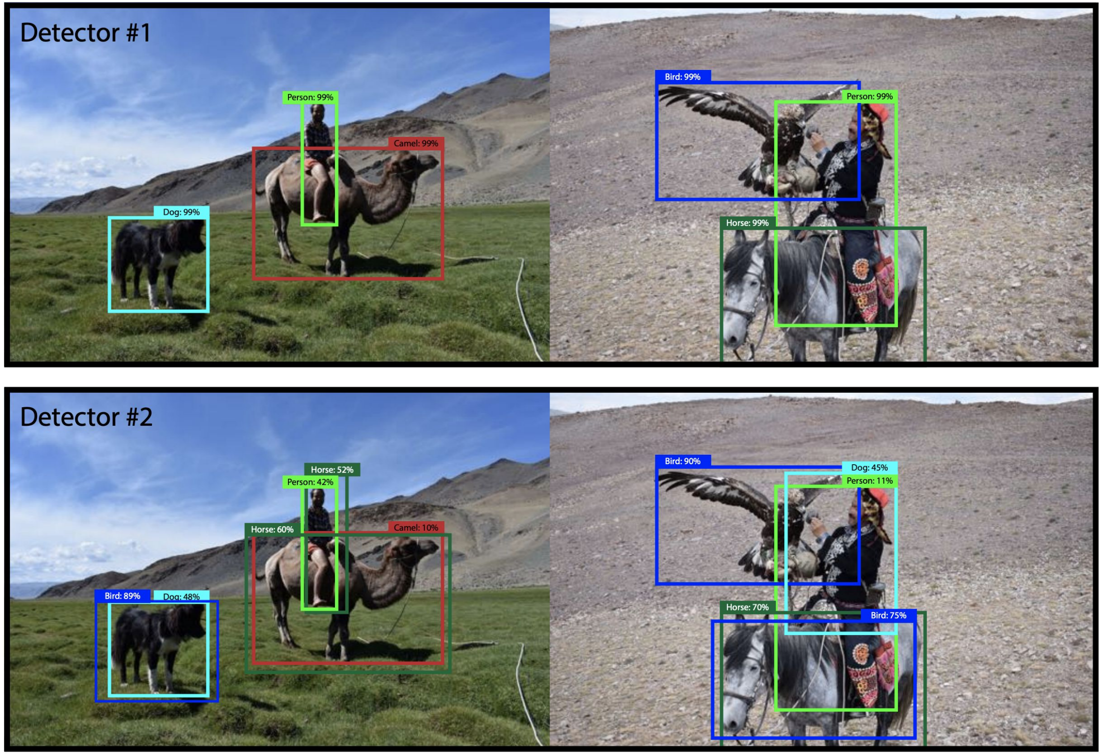
See https://medium.com/@jonathan_hui/map-mean-average-precision-for-object-detection-45c121a31173

mean Average Precision (mAP)

Problems

These two hypothetical detectors are perfect according to mAP over these two images. They are both perfect. Totally equal

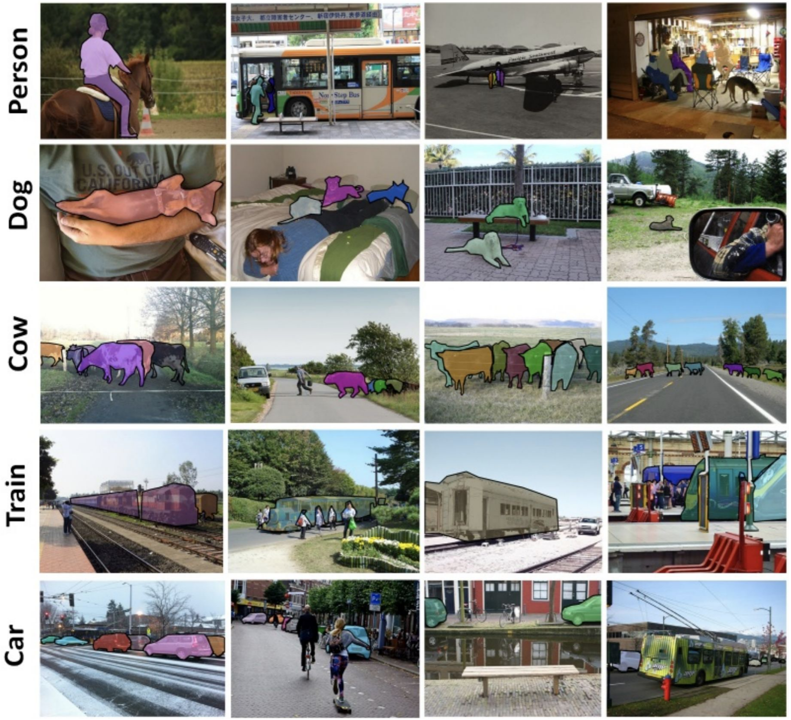
(from YOLOv3 article)

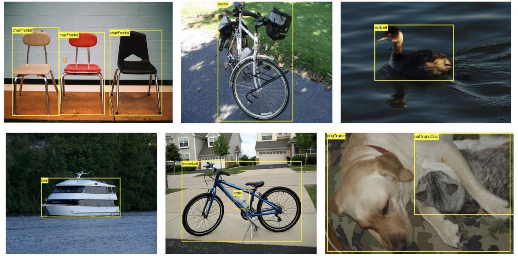


Datasets

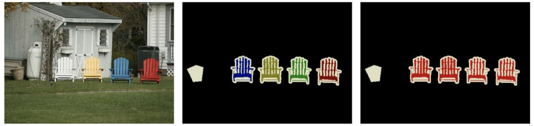
Detection datasets

1. PASCAL VOC 2012
2. ImageNet
3. COCO 2017
4. Open Images

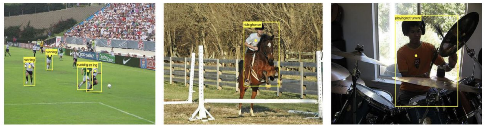




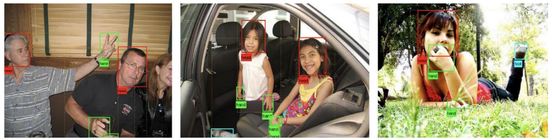
(a) Classification and detection



(b) Segmentation



(c) Action classification



(d) Person layout

Thanks to Ilya Zakharkin
for materials

PASCAL VOC 2012

Table 1 The VOC classes

Vehicles	Household	Animals	Other
Aeroplane	Bottle	Bird	Person
Bicycle	Chair	Cat	
Boat	Dining table	Cow	
Bus	Potted plant	Dog	
Car	Sofa	Horse	
Motorbike	TV/Monitor	Sheep	
Train			

The classes can be considered in a notional taxonomy

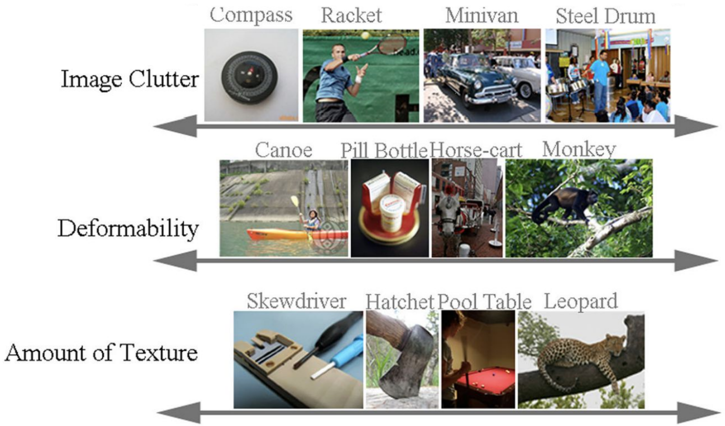
Relevant dataset overview:

http://host.robots.ox.ac.uk/pascal/VOC/pubs/everingham15.pdf

- PASCAL Visual Object Classes
- developed since 2005 and 4 classes
- 11,5k images
- 20 classes
- 31,5k detections
- 7k segmentations
- Images came from flickr.com

ImageNet

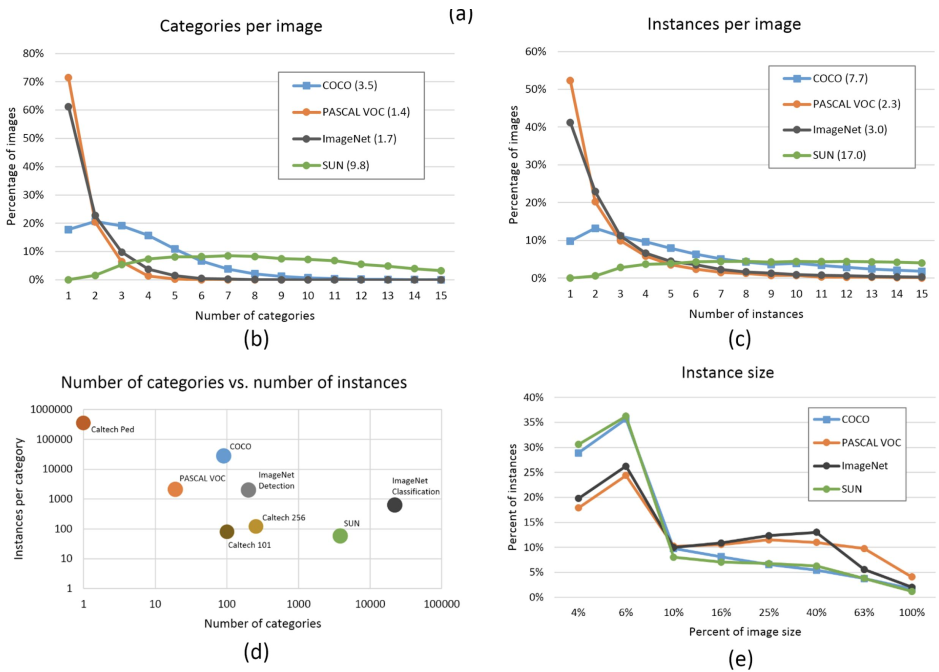
- organized according to the WordNet hierarchy
- 14.2m images
- 1000 classes
- 1m images with BBs
- more fine graded classes than PASCAL
- more diverse objects properties



http://image-net.org/explore

MS COCO 2017

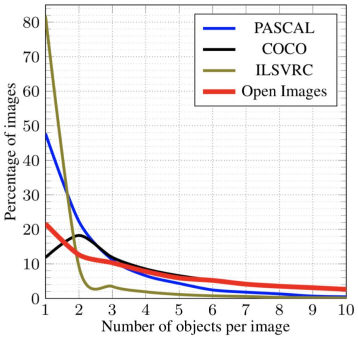
COCO
Common Objects in Context

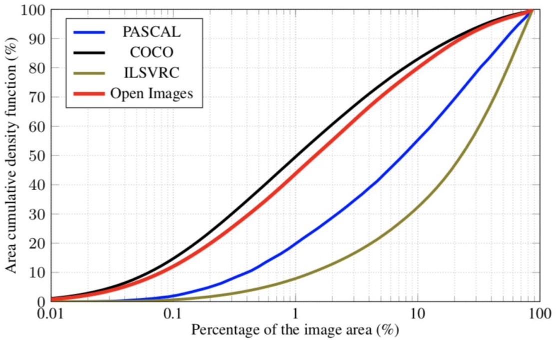


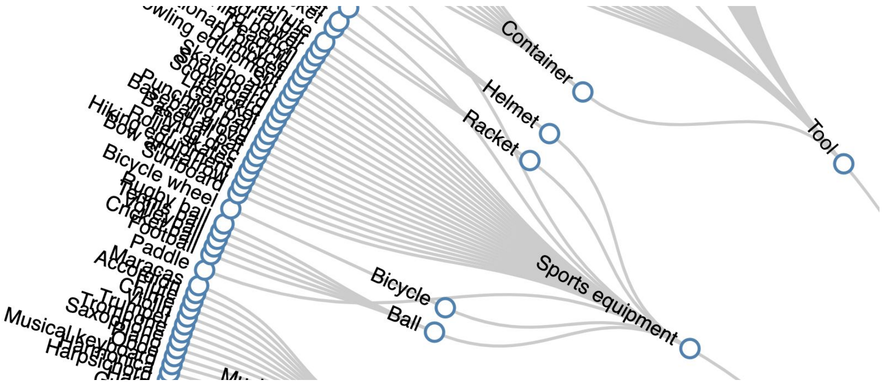
- MicroSoft Common Objects in COntext
- 328k images
- 91 object categories
- 82 having more than 5,000 labeled instances
- 2,5m labeled instances in total
- 250k people with keypoints
- online exploration tool http://cocodataset.org/#explore
- overview https://arxiv.org/pdf/1405.0312.pdf

Open Images

- Provided up to 2020 (v6)
- 15m boxes on 600 categories
- 2,8m instance segmentations on 350 categories
- 36,5m image-level labels on 20k categories
- 391k relationship annotations of 329 relationships
- Extension - 478k crowdsourced images with 6k+ categories







https://storage.googleapis.com/openimages/web/factsfigures.html

Practical notice

Mixup your small datasets with opensourse dataset
to prevent overfitting

R-CNN

Detection as Classification

CAT? NO

DOG? NO

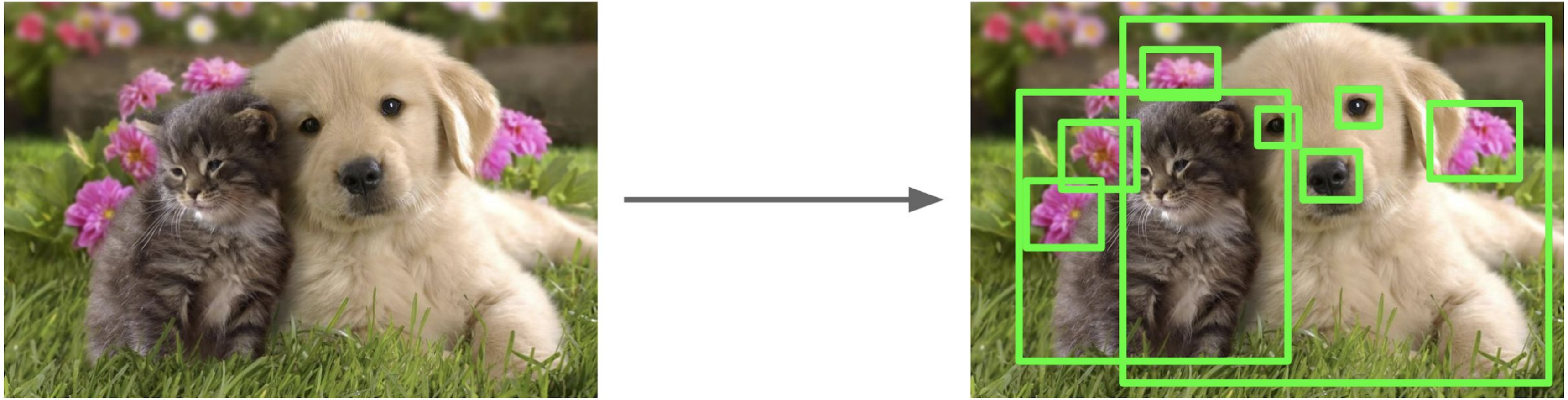
Detection as Classification

CAT? YES!

DOG? NO

Region Proposals

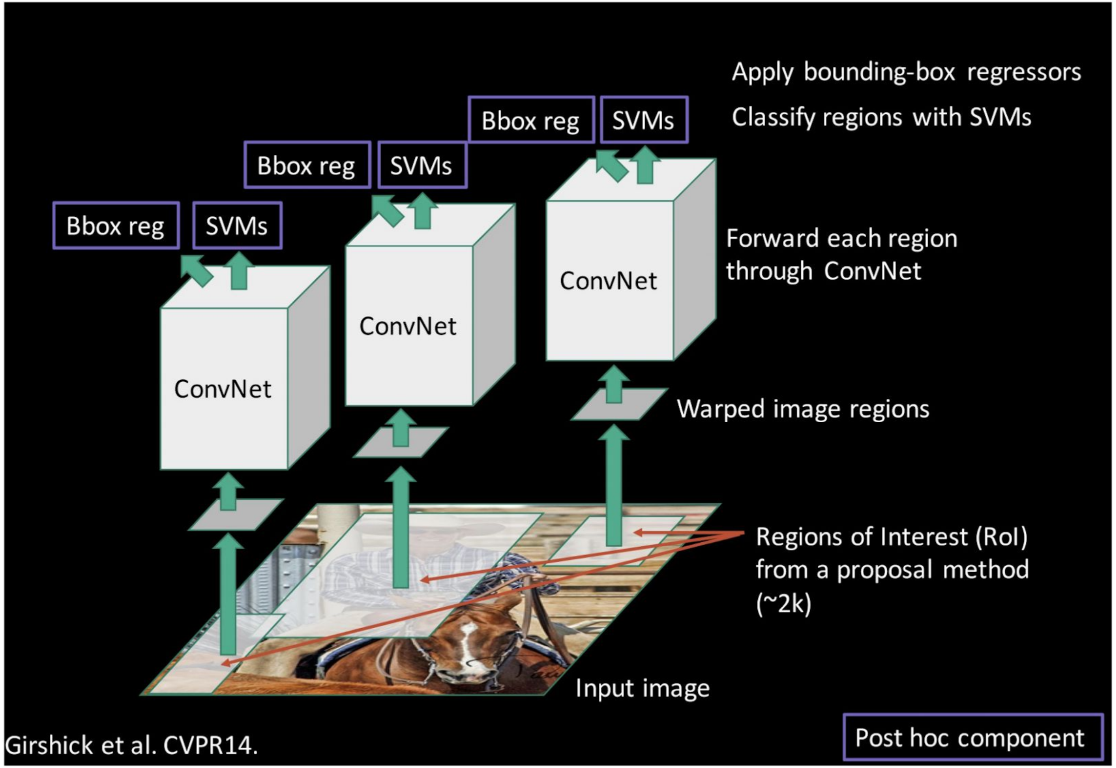
- Find “blobby” image regions that are likely to contain objects
- “Class-agnostic” object detector
- Look for “blob-like” regions



Region Proposals: Many other choices

Method	Approach	Outputs Segments	Outputs Score	Control #proposals	Time (sec.)	Repea-tability	Recall Results	Detection Results
Bing [18]	Window scoring		✓	✓	0.2	★★★	★	·
CPMC [19]	Grouping	✓	✓	✓	250	-	★★	★
EdgeBoxes [20]	Window scoring		✓	✓	0.3	★★	★★★	★★★
Endres [21]	Grouping	✓	✓	✓	100	-	★★★	★★
Geodesic [22]	Grouping	✓		✓	1	★	★★★	★★
MCG [23]	Grouping	✓	✓	✓	30	★	★★★	★★★
Objectness [24]	Window scoring		✓	✓	3	·	★	·
Rahtu [25]	Window scoring		✓	✓	3	·	·	★
RandomizedPrim's [26]	Grouping	✓		✓	1	★	★	★★
Rantalankila [27]	Grouping	✓		✓	10	★★	·	★★
Rigor [28]	Grouping	✓		✓	10	★	★★	★★
SelectiveSearch [29]	Grouping	✓	✓	✓	10	★★	★★★	★★★
Gaussian				✓	0	·	·	★
SlidingWindow				✓	0	★★★	·	·
Superpixels		✓			1	★	·	·
Uniform				✓	0	·	·	·

Putting it together: R-CNN

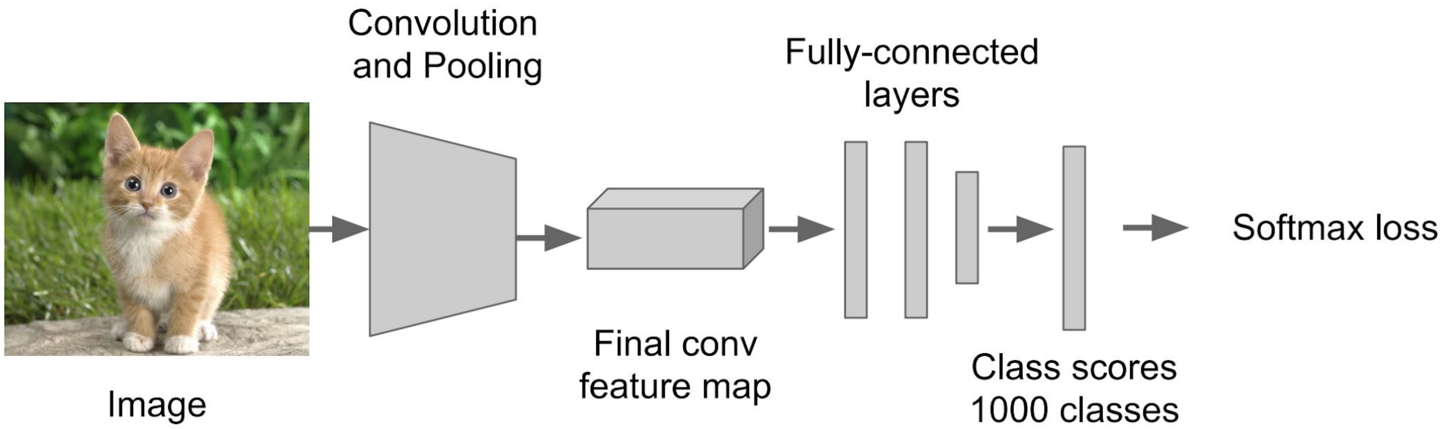


Girschick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014

Slide credit: Ross Girschick

R-CNN Training

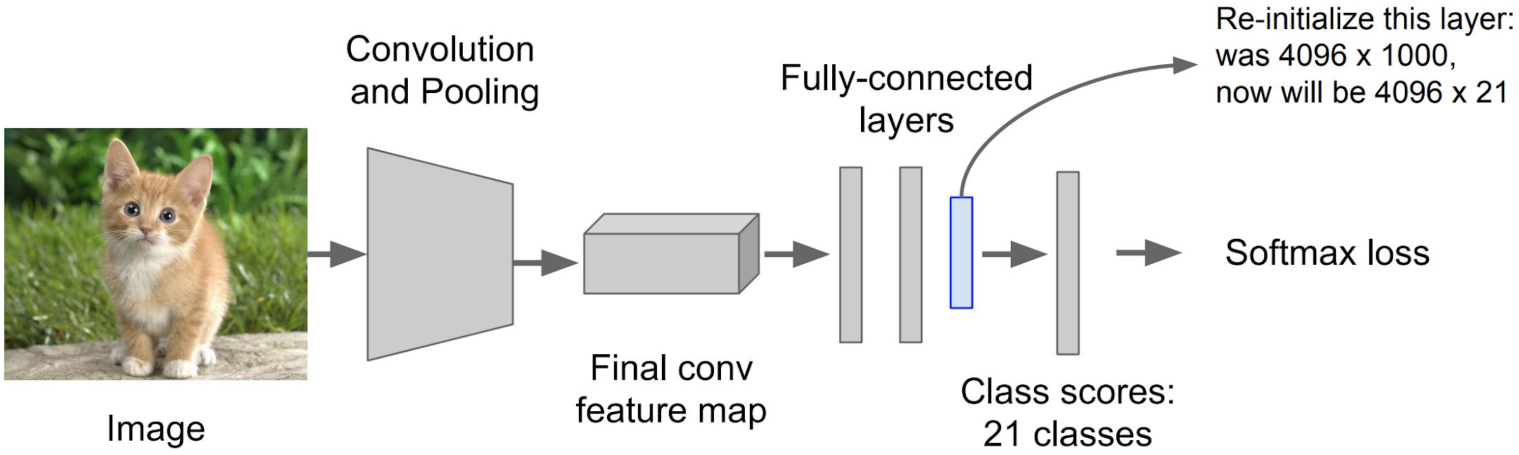
Step 1: Train (or download) a classification model for ImageNet (AlexNet)



R-CNN Training

Step 2: Fine-tune model for detection

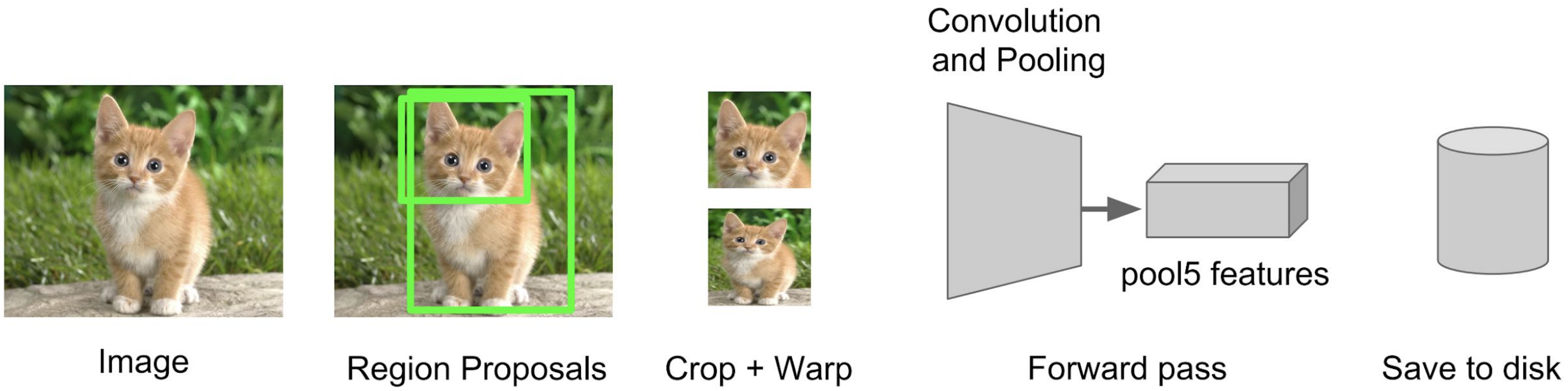
- Instead of 1000 ImageNet classes, want 20 object classes + background
- Throw away final fully-connected layer, reinitialize from scratch
- Keep training model using positive / negative regions from detection images



R-CNN Training

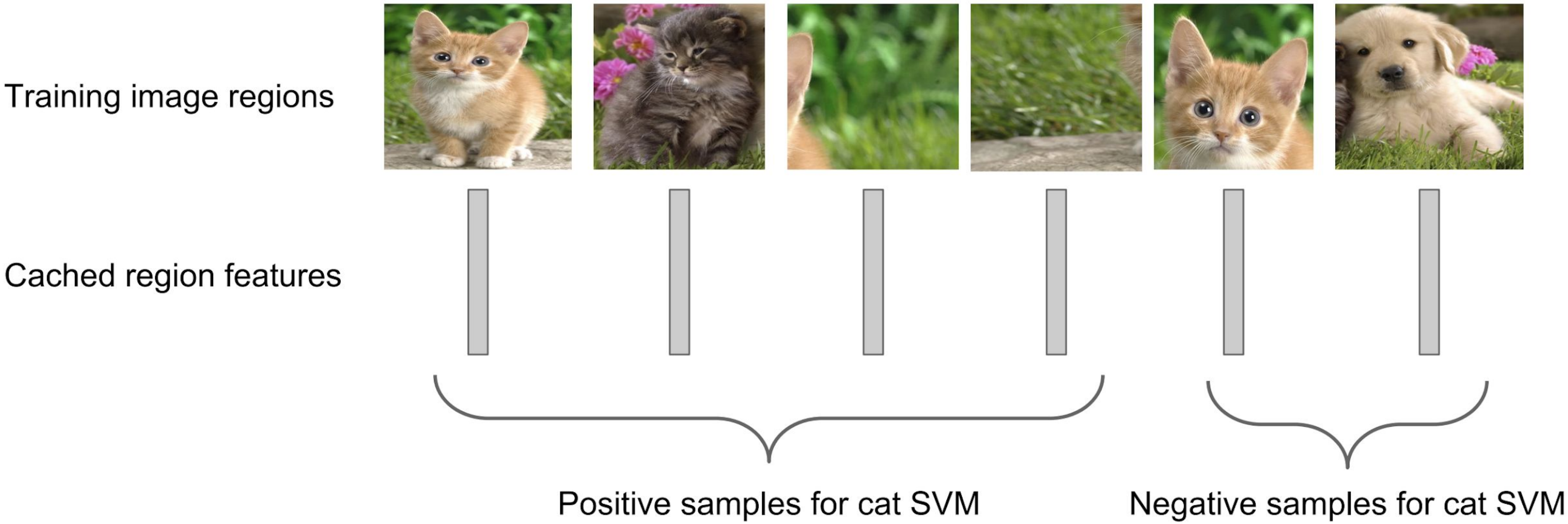
Step 3: Extract features

- Extract region proposals for all images
- For each region: warp to CNN input size, run forward through CNN, save pool5 features to disk
- Have a big hard drive: features are ~200GB for PASCAL dataset!



R-CNN Training

Step 4: Train one binary SVM per class to classify region features

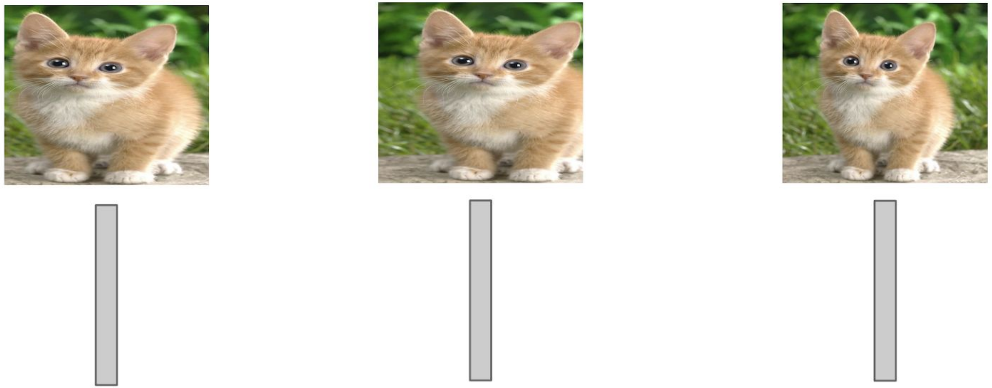


R-CNN Training

Step 5 (bbox regression): For each class, train a linear regression model to map from cached features to offsets to GT boxes to make up for "slightly wrong" proposals

Training image regions

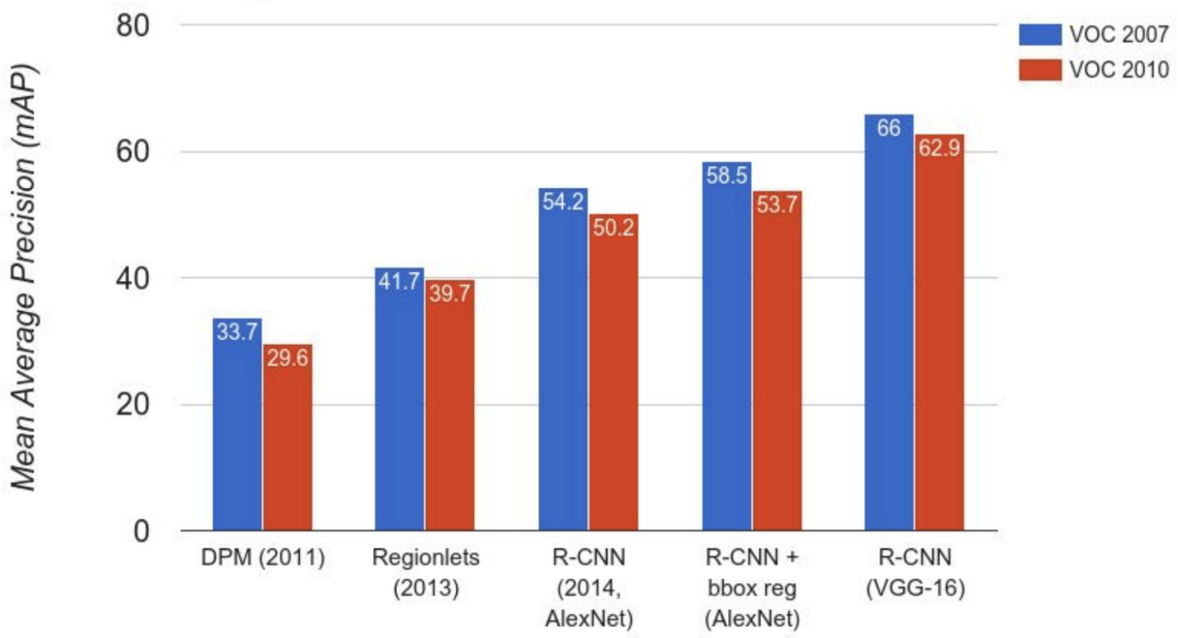
Cached region features



Regression targets
(dx, dy, dw, dh)
Normalized coordinates

(0, 0, 0, 0) Proposal is good	(.25, 0, 0, 0) Proposal too far to left	(0, 0, -0.125, 0) Proposal too wide

R-CNN Results



Wang et al, "Regionlets for Generic Object Detection", ICCV 2013

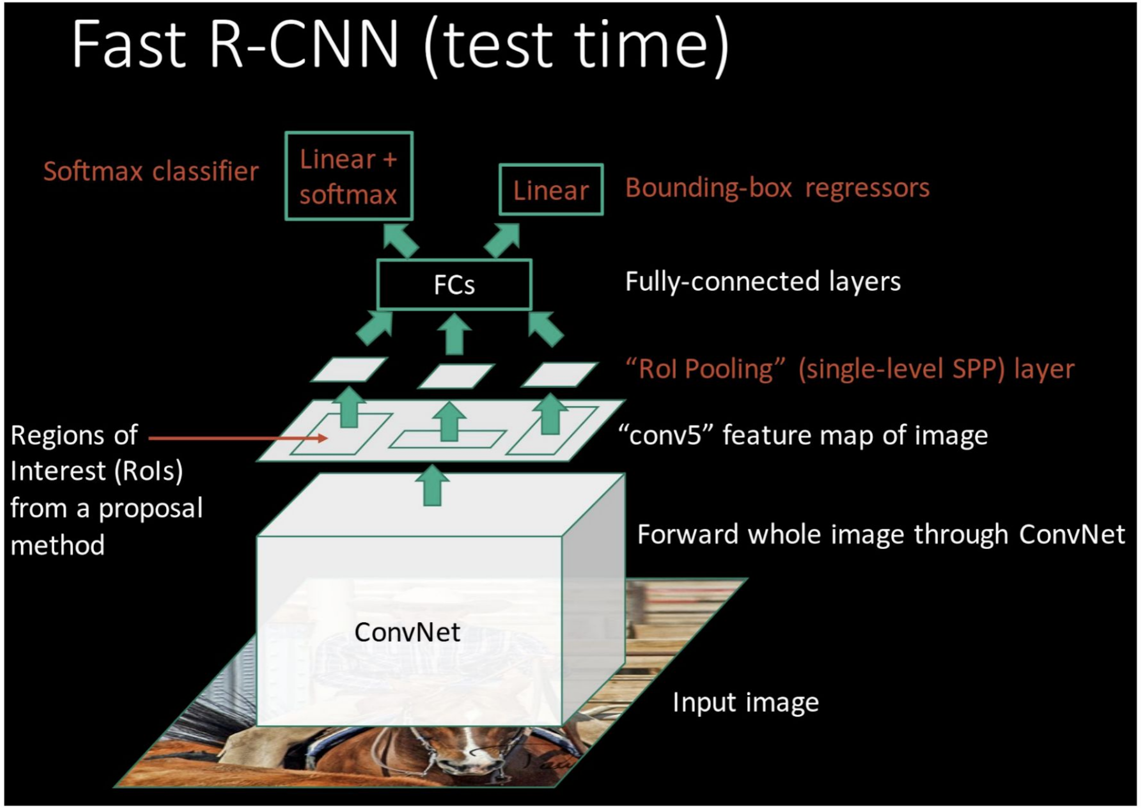
R-CNN Problems

1. Slow at test-time: need to run full forward pass of CNN for each region proposal

2. SVMs and regressors are post-hoc: CNN features not updated in response to SVMs and regressors

3. Complex multistage training pipeline

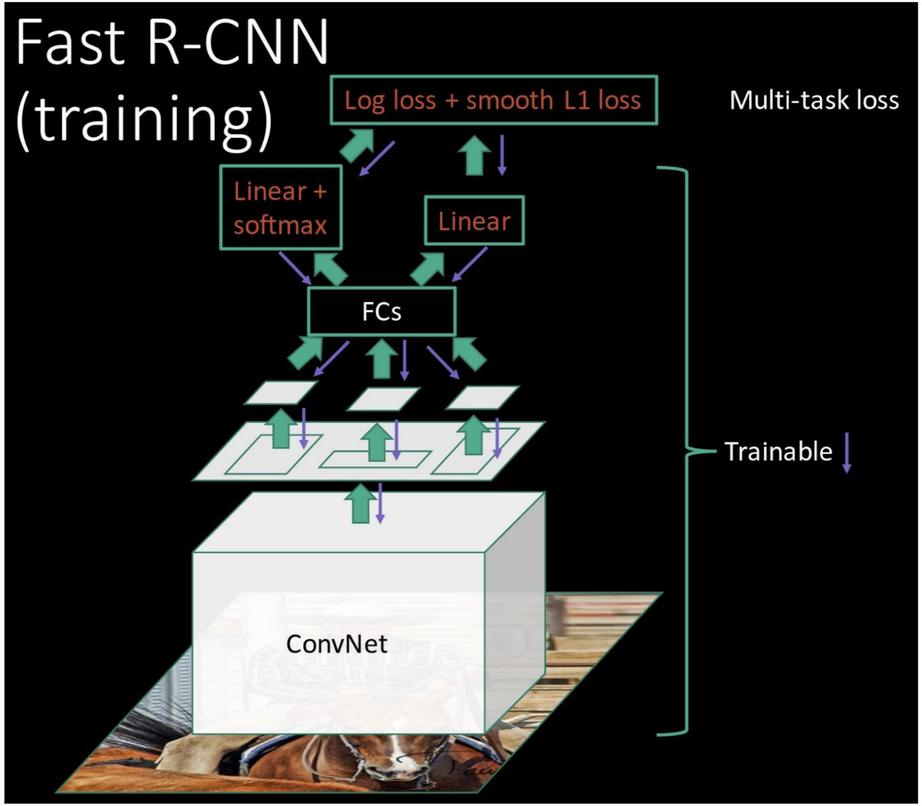
Fast R–CNN



R-CNN Problem #1:
Slow at test-time due to independent forward passes of the CNN

Solution:
Share computation of convolutional layers between proposals for an image

Fast R–CNN



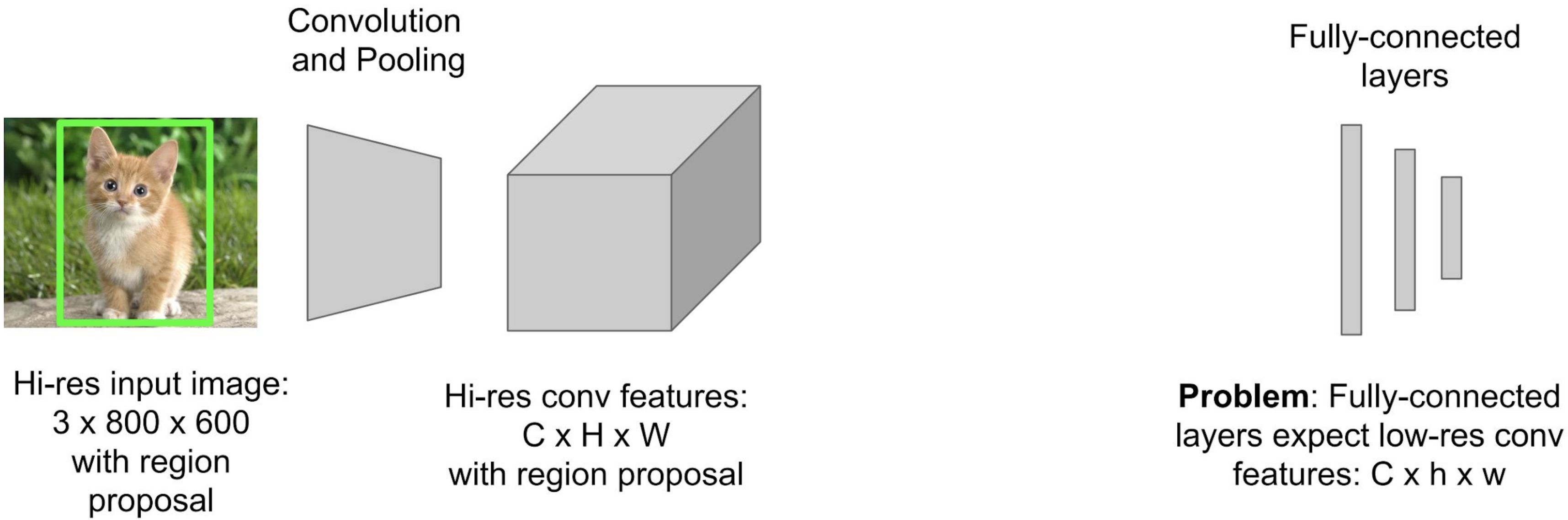
R-CNN Problem #2:
Post-hoc training: CNN not updated in response to final classifiers and regressors

R-CNN Problem #3:
Complex training pipeline

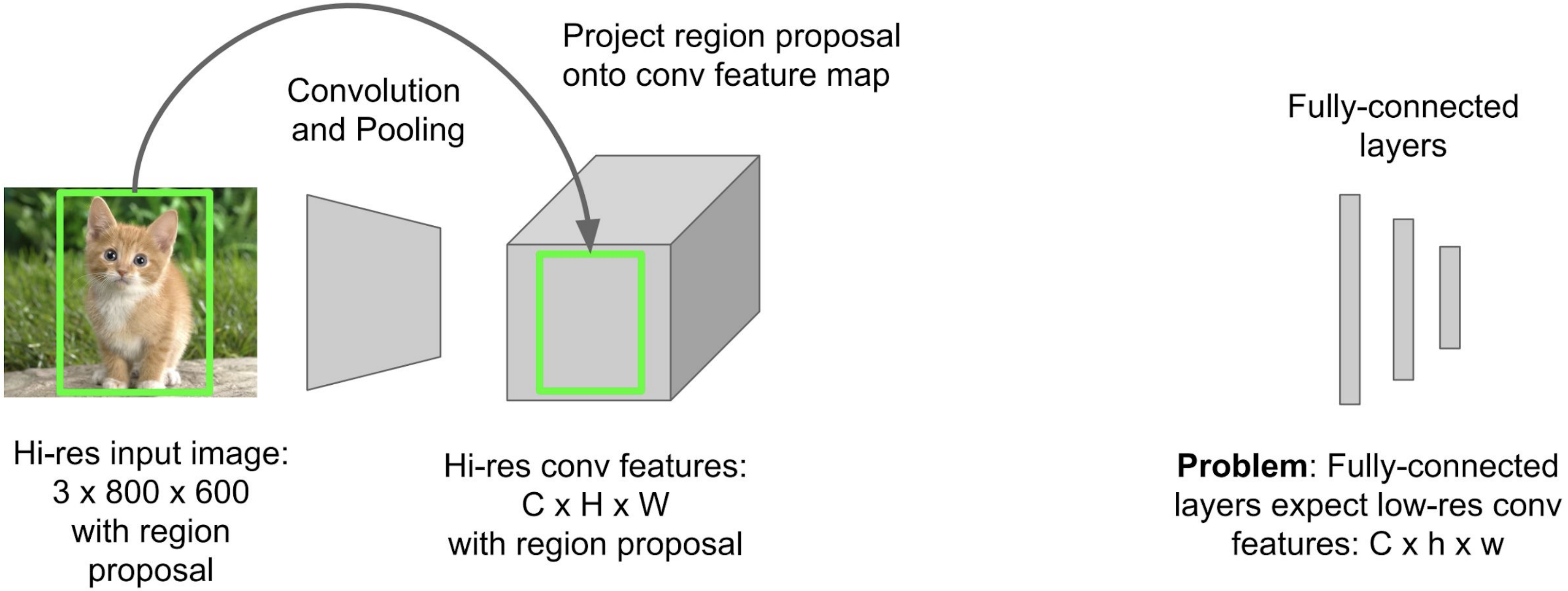
Solution:
Just train the whole system end-to-end all at once!

Slide credit: Ross Girschick

Fast R-CNN: Region of Interest Pooling



Fast R-CNN: Region of Interest Pooling



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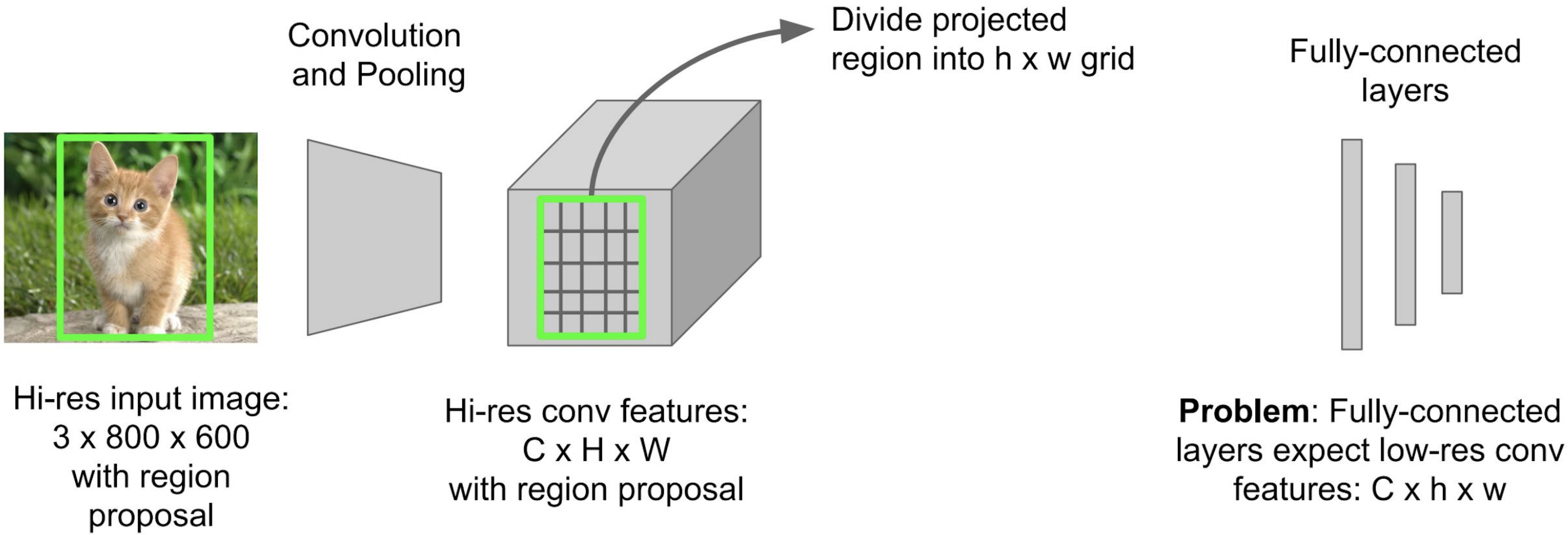
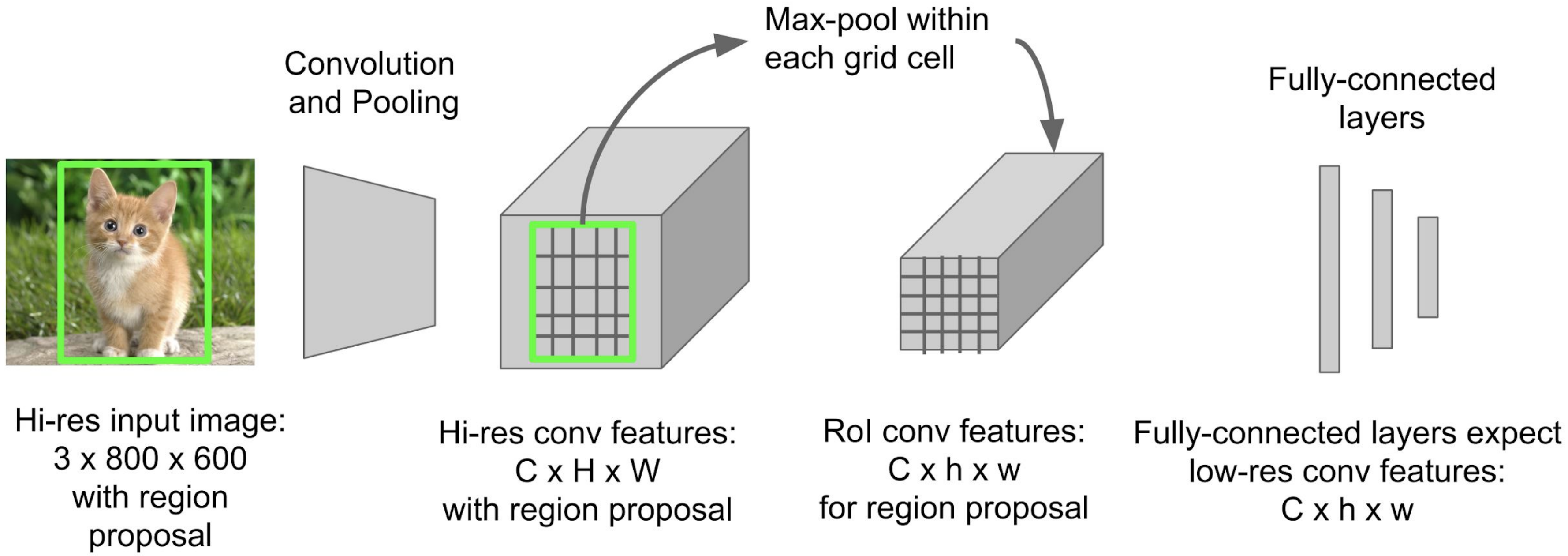
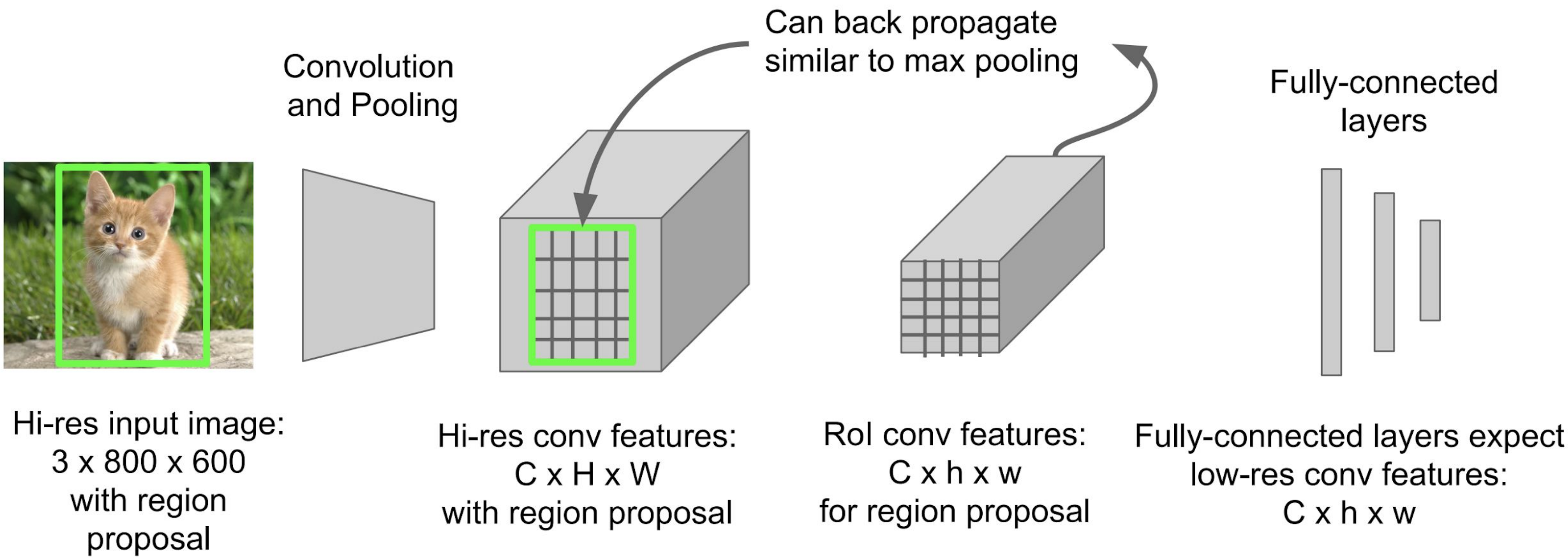


Fast R-CNN: Region of Interest Pooling



Fast R-CNN: Region of Interest Pooling



Fast R-CNN Results

		R-CNN	Fast R-CNN
Faster!	Training Time:	84 hours	**9.5 hours**
	(Speedup)	1x	**8.8x**
FASTER!	Test time per image	47 seconds	**0.32 seconds**
	(Speedup)	1x	**146x**
Better!	mAP (VOC 2007)	66.0	**66.9**

Using VGG-16 CNN on Pascal VOC 2007 dataset

Fast R-CNN Problem

Test-time speeds don't include region proposals

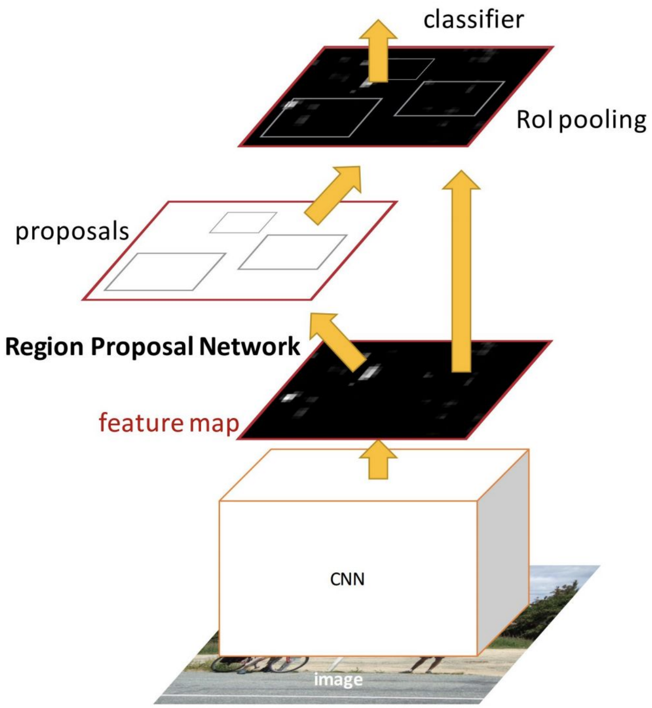
	R-CNN	Fast R-CNN
Test time per image	47 seconds	**0.32 seconds**
(Speedup)	1x	**146x**
Test time per image with Selective Search	50 seconds	**2 seconds**
(Speedup)	1x	**25x**

Fast R-CNN Problem Solution

Test-time speeds don't include region proposals
Just make the CNN do region proposals too!

	R-CNN	**Fast R-CNN**
Test time per image	47 seconds	**0.32 seconds**
(Speedup)	1x	**146x**
Test time per image with Selective Search	50 seconds	**2 seconds**
(Speedup)	1x	**25x**

Faster R-CNN



Insert a **Region Proposal Network (RPN)** after the last convolutional layer

RPN trained to produce region proposals directly; no need for external region proposals!

After RPN, use RoI Pooling and an upstream classifier and bbox regressor just like Fast R-CNN

Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015

Slide credit: Ross Girschick

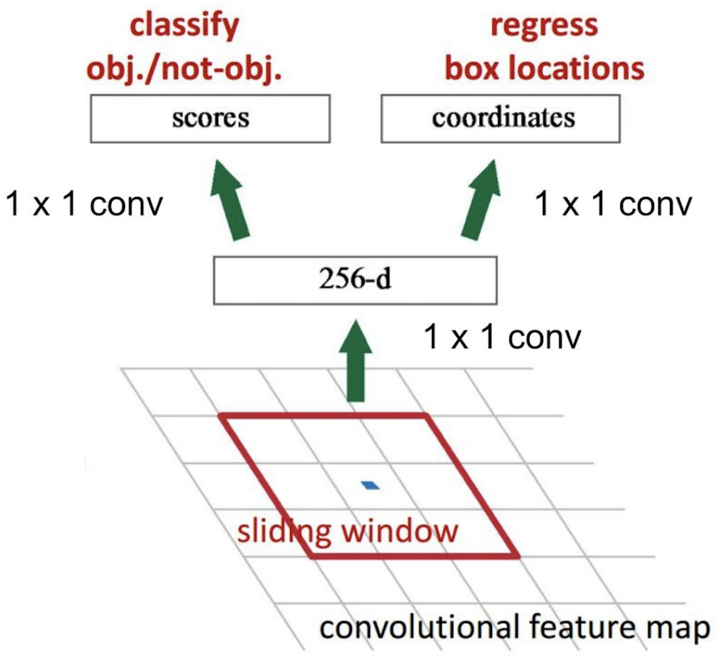
Faster R-CNN: Region Proposal Network

Slide a small window on the feature map

Build a small network for:
- classifying object or not-object, and
- regressing bbox locations

Position of the sliding window provides localization information with reference to the image

Box regression provides finer localization information with reference to this sliding window



Slide credit: Kaiming He

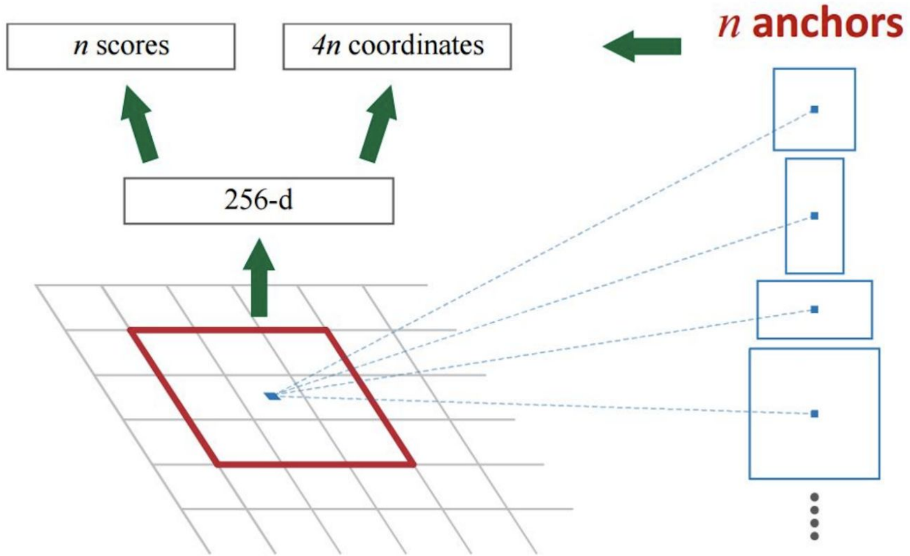
Faster R-CNN: Region Proposal Network

Use **N anchor boxes** at each location

Anchors are **translation invariant**: use the same ones at every location

Regression gives offsets from anchor boxes

Classification gives the probability that each (regressed) anchor shows an object



Faster R-CNN: Training

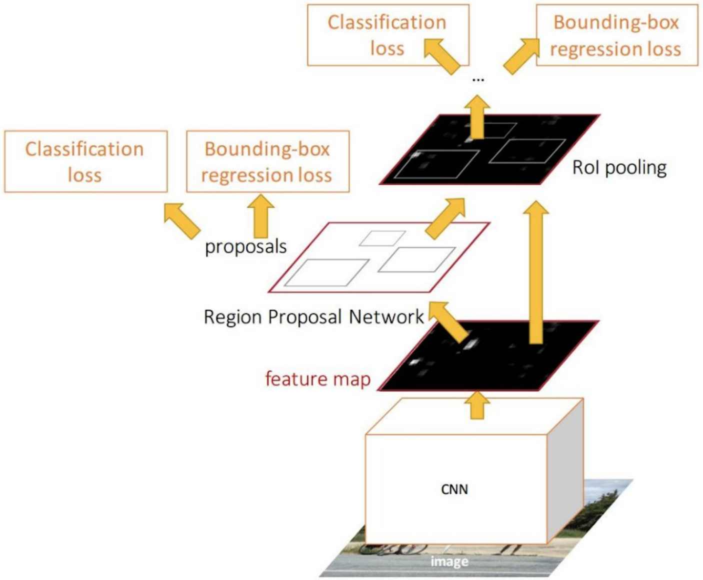
In the paper: Ugly pipeline

- Use alternating optimization to train RPN, then Fast R-CNN with RPN proposals, etc.
- More complex than it has to be

Since publication: Joint training!

One network, four losses

- RPN classification (anchor good / bad)
- RPN regression (anchor -> proposal)
- Fast R-CNN classification (over classes)
- Fast R-CNN regression (proposal -> box)

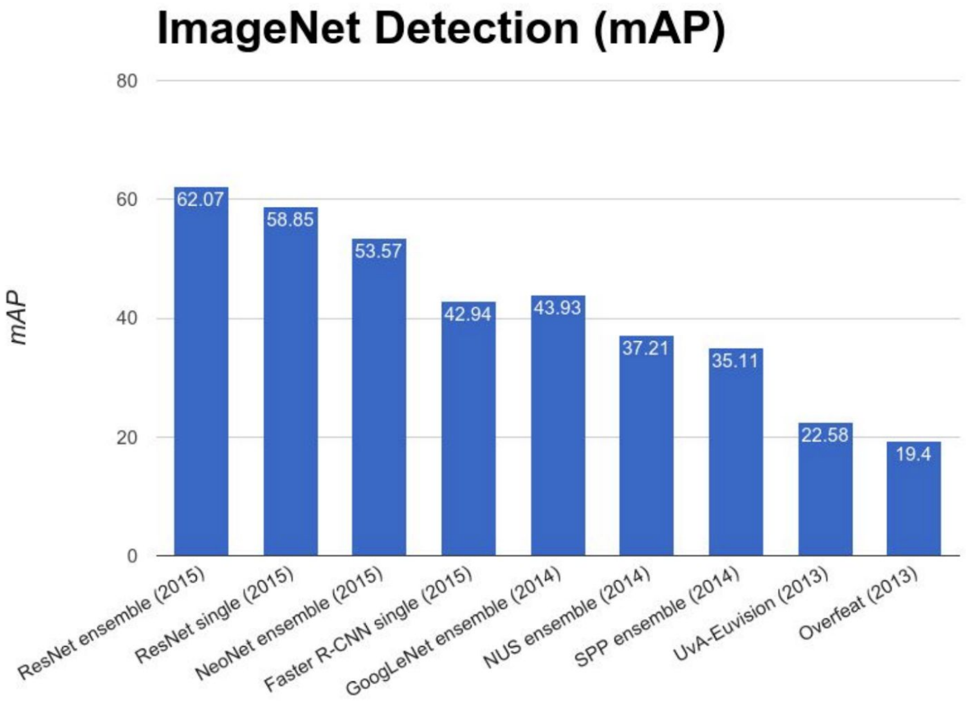


Slide credit: Ross Girschick

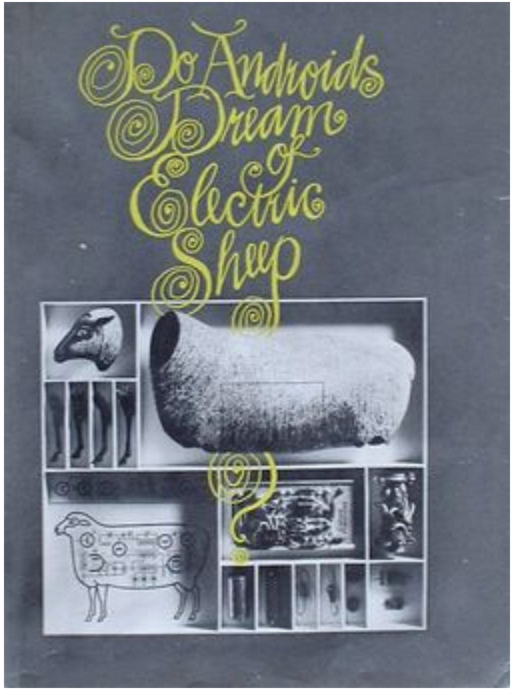
Faster R-CNN: Results

	R-CNN	**Fast R-CNN**	**Faster R-CNN**
Test time per image (with proposals)	50 seconds	2 seconds	**0.2 seconds**
(Speedup)	1x	25x	**250x**
mAP (VOC 2007)	66.0	**66.9**	**66.9**

ImageNet Detection 2013 – 2015



More CV to come...



Next time will:

- try to Only Look Once at an image
- know Do Androids Dream of Electric Sheep?